Credit Name: CSE 2110 Procedural Programming 1
Assignment Name: Chapter 5 Mastery Exercise 6 (DigitsSum)

The following questions can help you in thinking critically about your problem-solving processes:

Understanding the Problem
How did you approach understanding the challenge?
Working through the IPO chart before I started coding was how I understood the main steps of the challenge. I would ask for an integer, break it down into its digits, and then sum them up. The sum would then be displayed to the user
Were there any parts of the problem you found confusing at first? If so, how did you resolve that confusion?
I understood that dividing by ten and taking the remainder would consistently get me the ones digit of the number, but I wasn't sure of the best way to remove that digit from the number once I found it. Eventually I realized that because java automatically truncates any decimals after division, I could just divide the number by 10 when I wanted to move on to the next digit. To get through all the digits I would need a while loop. For that, I used the textbook as a

Planning the Solution
Did you create a plan or break the problem into smaller steps before coding?
After practicing with the skill builders I felt quite confident with the main code, but what I did do was test each calculation in regards to the digits and sum to make sure it would work once implemented. I also tested each if else statement seperatly from the rest, to check it for obvious errors.
How did you decide on the tools, data structures, or algorithms to use?
I used integer variables because the numbers had to be positive integers. I used the while loop so that I could keep finding the digits of a number until there are none left. If else statements helped allow the code to give different answers depending the number input.

Implementation
Did you write the code in small pieces or attempt the entire solution at once?
While I was coding I would test small sections such as calculations, the while loop, if else statments... Just to make sure all the parts were working properly for when I put it all together.
How did you test your solution along the way to make sure it was working?
I would just run a few lines of test code (or the old code, plus the new lines I just added) to see if it worked. This would have helped me catch any errors early and be able to fix them without entirely rewriting my code.

Overcoming Challenges
What part of the problem was the most difficult for you?
Figuring out how to use the while loop to help find each digit until none are left.
How did you handle moments when you felt stuck or unsure of what to do next?
I asked some friends what they would do or I would refer to the latest chapters in the textbook.

Learning
Was there anything you learned that you think will help you with future challenges?
In this code I learned some applications javas division as well as the usefulness of a while loop nested in an if else statement, and how to use the the while loop to make calculate and display multiple similiar things.